

Algebra PhD Exam, May 2013

Answer seven problems. Write your answers clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

- Let K be a field and let $f(x) \in K[x]$ be a non-zero polynomial.
 - Define what is a splitting field for $f(x)$ over K .
 - If F_1 and F_2 are splitting fields for $f(x)$ over K prove, from your definition, that F_1 and F_2 are isomorphic fields.
- Let $F = \mathbb{Q}(\sqrt{3}, \sqrt{5})$ be the smallest subfield of $\mathbb{C}$ which contains $\sqrt{3}$ and $\sqrt{5}$. How many subfields does F have?
- Let A and B be finite abelian groups and suppose that $|A|$ and $|B|$ are relatively prime. Let $T = A \otimes_{\mathbb{Z}} B$. Show that $|T| = 1$.
- Let C be a category and let $(A_i)_{i \in I}$ be a family of objects of C .
 - Define what is a coproduct of $(A_i)_{i \in I}$ in C .
 - Prove that any two coproducts of $(A_i)_{i \in I}$ in C are equivalent (some texts would say isomorphic) in C .
- Let R be a ring.
 - Define what it means for an R -module M to be projective.
 - Prove that if
$$0 \rightarrow A \rightarrow B \rightarrow P \rightarrow 0$$
is an exact sequence of R -modules and P is projective then the exact sequence splits.
- Let F be a free group with free generators a and b . Let N be the smallest normal subgroup of F such that $a^2 \in N$ and $b^2 \in N$. Prove that F/N is infinite.
- State and prove the Lying-Over Theorem on prime ideals and ring extensions.
- Prove the following version of Nakayama's Lemma. Let A be a commutative ring with identity, let M be a finitely generated A -module, and let I be an ideal contained in the intersection of all the maximal ideals of A . Show that if $IM = M$ then $M = \{0\}$.
- An invertible ideal in an integral domain that is a local ring is principal.
- Let $\overline{\mathbb{Q}}$ be an algebraic closure of $\mathbb{Q}$ and $v \in \overline{\mathbb{Q}}$ with $v \notin \mathbb{Q}$. Let E be a subfield of $\overline{\mathbb{Q}}$ maximal with respect to the condition that $v \notin E$. Prove that every finite-dimensional extension of E is cyclic.
- Classify (up to ring isomorphism) all semisimple rings of order 720.